A tropical formula for non-archimedean local heights (joint work with José Burgos and Walter Gubler)

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- **José Burgos, Walter Gubler, K.K.: A tropical formula for non-archimedean local heights.** arXiv:2512.07431.
- K field complete with respect to non-trivial non-archimedean absolute value $|\ |_K$.
- For simplicity, we assume that K is algebraically closed (for example $K = \mathbb{C}_p$).
- $K^\circ := \{x \in K \mid |x|_K \leq 1\}$ the valuation ring,
- $K^{\circ\circ} := \{x \in K \mid |x|_K < 1\}$ the maximal ideal, and
- $\widetilde{K} = K^\circ/K^{\circ\circ}$ residue field.
- X d-dimensional K-variety, i.e. integral K-scheme separated and of finite type.
- **General Aim.** Give analytic description of non-archimedean local heights over K, i.e. of arithmetic intersection numbers of arithmetic divisors on proper flat models $\mathscr{X}$ of X over Spec K°.

- $X^{\mathrm{an}} :=$ **Berkovich analytification of** X. Consists of pairs $(p, |\ |_p)$ where $p \in X$ and $|\ |_p$ is absolute value on $\kappa(p) = \mathcal{O}_{X,p}/\mathfrak{m}_{X,p}$ which extends $|\ |_K$. Equip X^{an} with coarsest topology such that

$$\pi : X^{\mathrm{an}} \longrightarrow X,\ (p, |\ |_p) \longmapsto p$$

is continuous and for all $U \subseteq X$ Zariski open and $f \in \mathcal{O}_X(U)$

$$|f| : U^{\mathrm{an}} = \pi^{-1}(U) \longrightarrow \mathbb{R},\ (p, |\ |_p) \longmapsto |f(p)| := |f + \mathfrak{m}_{X,p}|_p$$

is continuous as well.

- Berkovich constructs non-archimedean analytic space $(X^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}})$ that is path connected, locally compact and Hausdorff.
- If $X = \mathrm{Spec}(A)$ for some K-algebra A of finite type, then $\mathrm{Spec}(A)^{\mathrm{an}}$ is the set

$$\{\text{multiplicative seminorms } |\ | : A \to \mathbb{R} \text{ such that } |\ ||_K = |\ |_K\}.$$

- Let $N \cong \mathbb{Z}^n$ be a free abelian group of finite rank n.
- Let $M := \mathrm{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$ and $N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R} = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{R})$.
- Let $T = \mathrm{Spec}\, K[M] \cong \mathrm{Spec}\, K[T_1^{\pm 1}, \ldots, T_n^{\pm 1}]$ be (split) multiplicative torus over K with character lattice M.
- **Tropicalization.** There is a natural continuous proper map

$$\begin{aligned} \mathrm{trop} : T^{\mathrm{an}} \cong (\mathbb{G}_m^r)^{\mathrm{an}} &\longrightarrow N_{\mathbb{R}} \cong \mathbb{R}^r, \\ p = |\ |_p &\longmapsto (m \mapsto -\log |\chi_m|_p) \\ &= (-\log |T_1|_p, \ldots, -\log |T_n|_p) \end{aligned}$$

  that admits continuous section $G : N_{\mathbb{R}} \to T^{\mathrm{an}}$ by Gauss norms.
- Let W be open in X^{an}. A **moment map** is morphism $\varphi : W \to T^{\mathrm{an}}$ for some split torus T as above.
- A **moment map** $\varphi : W \to T^{\mathrm{an}}$ induces a continuous map

$$\varphi_{\mathrm{trop}} : W^{\mathrm{an}} \longrightarrow T^{\mathrm{an}} \xrightarrow{\mathrm{trop}} N_{\mathbb{R}} \cong \mathbb{R}^r.$$

  Next we describe the structure of the image $\varphi_{\mathrm{trop}}(W)$.

- An **(integral, $\mathbb{R}$-affine) polyhedron** Δ in $N_{\mathbb{R}}$ is a finite intersection (use $\langle .,. \rangle : M_{\mathbb{R}} \times N_{\mathbb{R}} \to \mathbb{R}$)

$$\Delta := \cap_{i=1}^{N} \{ \omega \in N_{\mathbb{R}} \,|\, \langle u_i, \omega \rangle \geq \gamma_i \}, \quad u_i \in M,\ \gamma_i \in \mathbb{R}.$$

  A polyhedron Δ determines affine subspace $\mathbb{A}_\Delta$ and linear subspace $\mathbb{L}_\Delta$ in $N_{\mathbb{R}}$, and its faces (written $\Delta' \prec \Delta$).
- A **polyhedral complex** Σ in $N_{\mathbb{R}}$ consists of a finite set Σ of polyhedra in $N_{\mathbb{R}}$ such that
  (a) $\Delta \in \Sigma \ \Rightarrow$ every face of Δ is in Σ,
  (b) $\Delta, \Delta' \in \Sigma \ \Rightarrow \Delta \cap \Delta'$ is empty or a face of Δ and Δ'.
- Call $|\Sigma| = \cup_{\Delta \in \Sigma} \Delta$ the *support* of Σ. Call $f : P \to \mathbb{R}$ with $P \subseteq N_{\mathbb{R}}$ **piecewise smooth (short ps)** if $P = |\Sigma|$ for some polyhedral complex Σ and $f|_\Delta$ is smooth for all Δ in Σ.
- A **weighted polyhedral complex** (Σ, m) **of pure dimension d** consists of polyhedral complex Σ of pure dimension d and weight function $m : \Sigma_d := \{\Delta \in \Sigma \,|\, \dim \Delta = d\} \to \mathbb{Z}$.

- Given $\sigma \in \Sigma_d$ and $\tau \in \Sigma_{d-1}$ such that $\tau \prec \sigma$, we choose some $\omega_{\sigma,\tau} \in N_\sigma := N \cap \mathbb{L}_\sigma$ whose class in N_σ/N_τ is the generator pointing in the direction of σ.
- Weighted polyhedral complex (Σ, m) satisfies *balancing condition* if

$$\sum_{\substack{\sigma \in \Sigma_d \\ \tau \prec \sigma}} m(\sigma)\omega_{\sigma,\tau} \in \mathbb{L}_\tau$$

  for all $\tau \in \Sigma_{d-1}$. In this case (Σ, m) is called a **tropical cycle**.
- Identify tropical cycles modulo polyhedral subdivision (with induced weights).
- **Theorem (Bieri–Groves 1984, Speyer–Sturmfels 2005) .** Let $\varphi_{\mathrm{trop}} : X^{\mathrm{an}} \to N_{\mathbb{R}}$ be induced by a closed immersion $\varphi : X \to T$. Then $\varphi_{\mathrm{trop}}\left(X^{\mathrm{an}}\right) \subseteq N_{\mathbb{R}}$ is support of integral, Γ-affine polyhedral complex of pure dimension d and equipped with canonical weights, $\varphi_{\mathrm{trop}}\left(X^{\mathrm{an}}\right)$ becomes a tropical cycle.

- **Superforms after Lagerberg.** For $\Omega \subseteq N_{\mathbb{R}}$ open define

$$\begin{aligned} A^{p,q}(\Omega) &:= A^p(\Omega, \mathbb{R}) \otimes_{\mathcal{C}^\infty(\Omega,\mathbb{R})} A^q(\Omega, \mathbb{R}) \\ &= A^p(\Omega, \mathbb{R}) \otimes_{\mathbb{Z}} \wedge^q M = \wedge^p M \otimes_{\mathbb{Z}} A^q(\Omega, \mathbb{R}). \end{aligned}$$

- Get products and differentials $d' = d \otimes \mathrm{id}$ and $d'' = \mathrm{id} \otimes d$ of bidegree $(1,0)$ and $(0,1)$.
- Get sheaf of bigraded differential algebras $(A^{\cdot,\cdot}, d', d'')$ on $N_{\mathbb{R}}$.
- Write sections of the sheaves $A^{p,q}$ in coordinates $x_1, \ldots, x_n$ as

$$\alpha = \sum_{\#I=p, \#J=q} f_{IJ} d'x_I \wedge d''x_J$$

with multi-index notation for $I, J \subseteq \{1, \ldots, n\}$ and

$$\begin{aligned} d'\alpha &= \sum_{i=1}^{n} \sum_{I,J} \frac{\partial f_{IJ}}{\partial x_i} d'x_i \wedge d'x_I \wedge d''x_J, \\ d''\alpha &= \sum_{j=1}^{n} \sum_{I,J} \frac{\partial f_{IJ}}{\partial x_j} d''x_j \wedge d'x_I \wedge d''x_J. \end{aligned}$$

- **Currents** $D^{p,q}(\Omega)$ are continuous functionals $A_c^{n-p,n-q}(\Omega) \to \mathbb{R}$.
- Integration $\int_\Delta \alpha$ of $\alpha \in A_c^{n-d,n-d}(N_\mathbb{R})$ over d-codimensional polyhedron Δ is well defined (use the lattice structure given by N). Get **current of integration** $\delta_\Delta \in D^{d,d}(N_\mathbb{R})$, $\delta_\Delta(\alpha) = \int_\Delta \alpha$.
- Let Ω be an open subset of $N_\mathbb{R}$. A Lagerberg current $T \in D(\Omega)$ is called **polyhedral** if there exists finite families $(\Delta_i)_{i\in I}$ of integral $\mathbb{R}$-affine polyhedra Δ_i in $N_\mathbb{R}$ and $(\alpha_i)_{\in\in I}$ of Lagerberg forms $\alpha_i \in A(\Omega)$ such that in $D(\Omega)$, we have

$$T = \sum_{i\in i} \alpha_i \wedge \delta_{\Delta_i}.$$

- A weighted polyhedral complex (Σ, m) of pure codimension d gives polyhedral current $\delta_{(\Sigma,m)} = \sum_{\Delta\in\Sigma_{n-d}} m_\Delta \delta_\Delta \in D^{d,d}(N_\mathbb{R})$.
- Theorem of Stokes yields

(0.1) (Σ, m) tropical cycle $\Leftrightarrow d'(\delta_{(\Sigma,m)}) = 0 \Leftrightarrow d''(\delta_{(\Sigma,m)}) = 0.$

Chambert-Loir–Ducros 2012:

- Construct sheaf A of **real-valued somooth forms** on X^{an} (or any good K-analytic space of pure dimension with empty boundary).
- $(A^{\cdot,\cdot}, d', d'')$ is sheaf of bigraded differential $\mathbb{R}$-algebras.
- Each form α is locally '**tropicalized**' by moment map $\varphi : W \to T^{\mathrm{an}}$ and Lagerberg form α_{trop} on $N_{\mathbb{R}}$ such that 'formally'

$$\alpha = \varphi_{\mathrm{trop}}^*(\alpha_{\mathrm{trop}}).$$

- Construction compatible with refinements of moment maps.
- Pull-back of forms for morphisms of analytic spaces.
- Stone–Weierstraß: $A_c^{0,0}(W)$ is dense in $C_c^0(W)$.
- Partitions of unity exist on paracompact analytic spaces.

- Given K-analytic space Y and $y \in Y$, can define complete residue field $\mathscr{H}(y)$. Get field extension $\mathscr{H}(y)/K$ and set

$$d(y) := \mathrm{rk}_{\mathbb{Z}}(|\mathscr{H}(y)^{\times}|/|K^{\times}|) + \mathrm{trdeg}(\widetilde{\mathscr{H}(y)}/\widetilde{K}) \leq \dim(Y).$$

- If $\alpha \in A^{p,q}(X)$ and $y \in \mathrm{supp}(\alpha)$, then $d(y) \geq \max(p,q)$.
- If $\alpha \in A^{p,q}(X)$ and Z subvariety of X of dimension $m < \max(p,q)$, then $Z^{\mathrm{an}} \cap \mathrm{supp}(\alpha) = \emptyset$.
- For W open in X^{an}, there is well defined **integral**

$$\int_W : A_c^{d,d}(W) \longrightarrow \mathbb{R}, \ \alpha \longmapsto \int_W \alpha.$$

- Sections of $A^{0,0}$ are **smooth functions**. Call $f : W \to \mathbb{R}$ **piecewise smooth (short ps)** if it can be tropicalized by ps function on $N_{\mathbb{R}}$.

- Let L be a line bundle on X. A **continuous metric** $\| \ \|$ on L associates with every $s \in \Gamma(U, L)$ for $U \subseteq X$ open a continuous function $\|s\| \colon U^{\mathrm{an}} \to \mathbb{R}_{\geq 0}$ such that $\|f \cdot s\| = |f| \, \|s\|$ for all $f \in \mathcal{O}_X(U)$, $\|s\| > 0$ if s is a frame of L, and $\|s\|_{V^{\mathrm{an}}} = \|s|_V\|$ for $V \subseteq U$ open.
- A continuous metric $\| \ \|$ on L is called **smooth (resp. ps)** if the function $\|s\| \colon U^{\mathrm{an}} \to \mathbb{R}$ is smooth (resp. ps).
- If $\| \ \|$ is smooth metric on L, then there is unique **1-st Chern form**

$$c_1(L, \| \ \|) \in A^{1,1}(X^{\mathrm{an}})$$

such that

$$c_1(L, \| \ \|)|_{U^{\mathrm{an}}} = -d'd'' \log \|s\|$$

for every frame s of L over U open in X.

- **Currents** $D^{p,q}(W)$ are continuous functionals $A_c^{d-p,d-q}(W) \to \mathbb{R}$.
- If $\| \ \|$ is continuous metric on L, then one can still define the **1-st Chern current**

$$[c_1(L, \| \ \|)] \in D^{1,1}(X^{\mathrm{an}})$$

s.t. $[c_1(L, \| \ \|)]|_{U^{\mathrm{an}}} = -d'd''[\log \|s\|]$ for frame s of L over U open.
- Algebraic cycle $Z \in Z^p(X)$ gives **current of integration**

$$\delta_Z \in D^{p,p}(X^{\mathrm{an}}).$$

- **Poincaré–Lelong formula (Chambert-Loir–Ducros 2012).** For meromorphic section s of L with continuous metric $\| \ \|$,

$$-d'd''[\log \|s\|] + \delta_{\mathrm{div}(s)} = [c_1(L, \| \ \|)]$$

holds in $D^{1,1}(X^{\mathrm{an}})$.

- Let $\mathscr{X}$ be a **model of** X over K°, i.e. a normal proper flat scheme over $S := \operatorname{Spec} K^\circ$ with generic fibre X.
- Valuative criterion of properness gives **reduction map**

$$\operatorname{red}\colon X^{\mathrm{an}} \longrightarrow \mathscr{X}_s := \mathscr{X} \otimes_{K^\circ} \tilde{K}.$$

- Let $\mathscr{X}$ be model of X over K° and $\mathscr{L}$ line bundle on $\mathscr{X}$ with $\mathscr{L}|_X = L$. A pair $(\mathscr{X}, \mathscr{L})$ determines a unique continuous metric $\| \ \|_{\mathscr{L}}$ on L such that a frame s of $\mathscr{L}$ over $\mathscr{U} \subseteq \mathscr{X}$ open satisfies $\|s\|_{\mathscr{L}} = 1$ on $\operatorname{red}^{-1}(\mathscr{U}_s)$. Such a metric is called an **algebraic metric**.
- A continuous metric $\| \ \|$ on L is a **model metric** if some power $(L^{\otimes k}, \| \ \|^{\otimes k})$ is an algebraic metric.
- **Problem.** Model metrics are piecewise smooth (even piecewise affine) but in general not smooth.

Gubler–K. 2017.

- Introduce sheaves $(B^{\cdot,\cdot}, d', d'')$ of so-called δ**-forms** on X^{an}.
- δ-forms are currents locally 'tropicalized' by polyhedral currents

$$\sum_{i=1}^{m} \alpha_i \wedge \delta_{C_i}.$$

  for tropical cycles C_i on some $N_{\mathbb{R}}$.
- There are products, pullbacks, δ-currents $E^{p,q}$, Poincaré–Lelong formula for δ-currents, ... Products for example $\delta_{C_1} \wedge \delta_{C_2} = \delta_{C_1 \cdot C_2}$.
- Call piecewise smooth metric $\| \ \|$ δ**-metric** if $[c_1(L, \| \ \|)]$ in $D^{1,1}(X^{\mathrm{an}})$ is 'universally' representable by a δ-form $c_1(L, \| \ \|)$.
- Model metrics are δ-metrics.
- For δ-metrics can define $c_1(L_1, \| \ \|_1) \wedge \ldots \wedge c_1(L_r, \| \ \|_r)$.
- Get measures $c_1(L_1, \| \ \|_1) \wedge \ldots \wedge c_1(L_d, \| \ \|_d)$ on X^{an} (Monge–Ampère or Chambert-Loir measures).

Non-archimedean Arakelov theory (Gubler–K. 2017).

- Let X be proper K-variety. A δ-current $g_Z \in E^{p-1,p-1}(X^{\mathrm{an}})$ is called a **Green current for a cycle** $Z \in Z^p(X)$ if and only if

$$d'd''g_Z + \delta_Z = [\omega_Z]$$

  holds for some δ-form $\omega(Z, g_Z) := \omega_Z \in B^{p,p}(X^{\mathrm{an}})$.
- Let $Z \in Z^p(X)$ be a prime cycle with a Green current g_Z. If s is meromorphic section of line bundle $(L, \| \ \|)$ with δ-metric such that $D = \operatorname{div}(s)$ intersects Z properly, then $*$**-product**

$$g_D * g_Z := g_D \wedge \delta_Z - c_1(L, \| \quad \|) \wedge g_Z \in E^{p,p}(X^{\mathrm{an}})$$

  is a Green current for the intersection $D \cdot Z \in Z^{p+1}(X)$ with

$$\omega(D \cdot Z, g_D * g_Z) = \omega(D, g_D) \wedge \omega(Z, g_Z).$$

Analytic definition of non-archimedean local heights.
Let $D_0, \ldots, D_d$ be Cartier divisors intersecting properly on proper K-variety X. Let $\mathcal{O}(D_0), \ldots, \mathcal{O}(D_d)$ be equipped with δ-metrics $\| \ \|_i$, associated Green currents $g_{D_i} = [-\log \|s_{D_i}\|_i]$ and associated arithmetic divisors $\widehat{D}_i := (D_i, g_{D_i})$, then

$$\lambda_{\widehat{D}_0,\ldots,\widehat{D}_d}(X) := \langle g_{D_0} * \cdots * g_{D_d}, 1 \rangle$$

is called **non-archimedean local height of X w.r.t.** $\widehat{D}_0, \ldots, \widehat{D}_d$.

Theorem (Gubler–K. 2017)

If we use algebraic metrics on $\mathcal{O}(D_0), \ldots, \mathcal{O}(D_d)$, then $\lambda_{\widehat{D}_0,\ldots,\widehat{D}_d}(X)$ is the usual local height of X in arithmetic geometry given as intersection number of the Cartier divisors on a corresponding model $\mathscr{X}$ of X.

Non-archimedean Arakelov theory works for divisors.

Tropical improvements (Andreas Mihatsch Selecta Math. 2023).

- Omit rationality conditions on polyhedra using 'weights'.
- Work with locally finite polyhedral complexes to get sheaves.
- Define a sheaf of δ**-forms** on $\mathbb{R}^r$ by taking locally finite polyhedral currents $T = \sum_{i \in I} \alpha_i \wedge \delta_{\Delta_i}$ such that $d'T$ and $d''T$ are again polyhedral currents.
- A polyhedral current T as above is a δ-form if and only if the family $(\alpha, \Delta_i)_{i \in I}$ satisfies a **balancing condition**.
- Construct products and pull-back for δ-forms generalizing tropical intersection theory product $\delta_{C_1} \wedge \delta_{C_2} = \delta_{C_1 . C_2}$.
- Describe product of δ-forms by Fulton–Sturmfels **fan displacement rule** and by **reduction to the diagonal**.

Analytic improvements (Andreas Mihatsch Duke Math. J. 2024).

- Improved theory of δ-forms by Gubler–K. 2017.
- Arbitrary good K-analytic spaces without boundary.
- Mihatsch used his more general tropical theoy of δ-forms on $\mathbb{R}^r$.
- The δ-forms constructed by Mihatsch are polyhedral currents on skeleta in sense of Ducros.
- Products, pullback, Poincaré–Lelong equation, . . .
- More δ-forms such that $\{\delta\text{-metrics}\} = \{\text{ps-metrics}\}$ giving for example height formula for all ps-metrics.
- Canonical metrics of line bundles over abelian varieties A are piecewise smooth and hence δ-metrics.
- Computation of concrete arithmetic intersection numbers on Lubin–Tate spaces.

Extended tropicalization using tropical toric varieties.

- Let M free abelian group of rank $n < \infty$ and $N := \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$.
- Let $\Sigma = \{\sigma\}_{\sigma \in \Sigma}$ be **rational polyhedral fan** in $N_{\mathbb{R}} := N \otimes_{\mathbb{Z}} \mathbb{R}$ with support $|\Sigma|$. Assume for simplicity that Σ is smooth.
- Let $\mathbb{T} = \mathrm{Spec}\, K[M]$ denote the split torus with character lattice M.
- Σ determines non-singular toric variety X_Σ with dense torus $\mathbb{T}$.
- For cone $\sigma \in \Sigma$, consider real quotient vector space

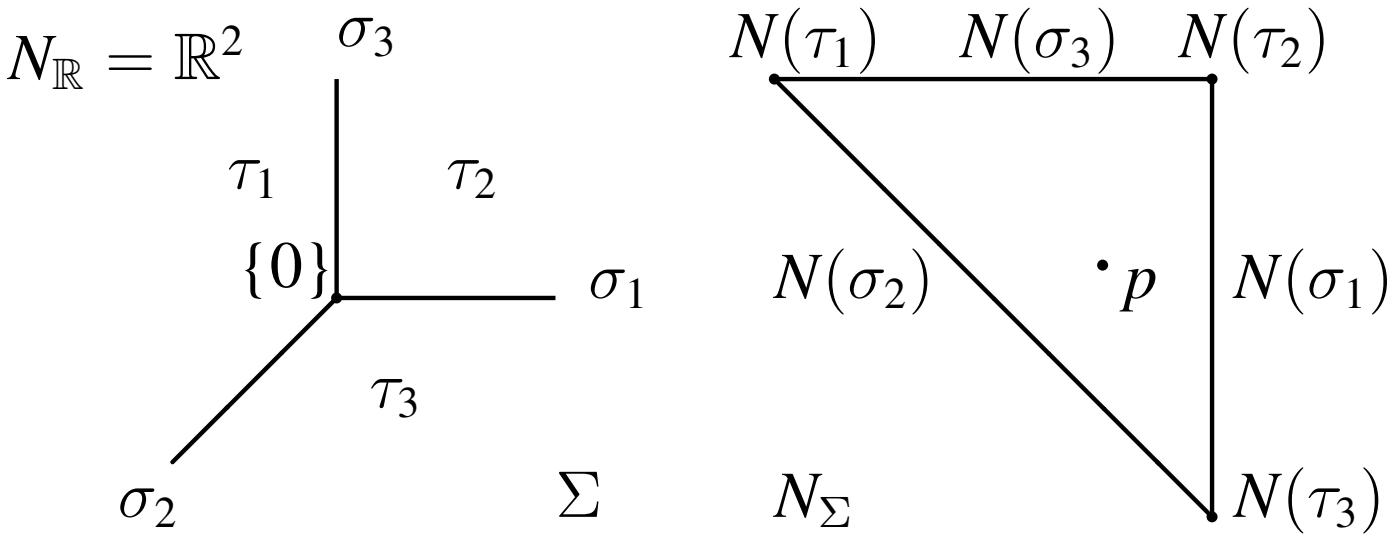
$$N(\sigma) := N_{\mathbb{R}} / \langle \sigma \rangle_{\mathbb{R}}.$$

- The *tropical toric variety* N_Σ associated with Σ is defined as

$$N_\Sigma := \coprod_{\sigma \in \Sigma} N(\sigma).$$

- If $\tau \prec \sigma$, then there is a natural projection $\pi_{\sigma,\tau} \colon N(\tau) \to N(\sigma)$.

- **Reminder:** The tropical projective plane $\mathbb{P}^2$



- Locally $(-\infty,\infty)^s \times [-\infty,\infty)^{n-s}$ with $[-\infty,\infty) \overset{\exp}{\to} [0,\infty) \subset \mathbb{R}$.
- N_Σ is Hausdorff, locally compact space with countable basis.
- For $p \in N_{\mathbb{R}}$ and $v \in |\Sigma|$ have $p + \infty v := \lim_{\mu\to\infty} p + \mu v \in N_\Sigma$.

Extended tropicalization

- $\mathbb{T}^{\mathrm{an}} = \{\text{multiplicative seminorms } |\ | \text{ on } K[M] \text{ with } |\ |\big|_K = |\ |_K\}$ and X_Σ^{an} the non-archimedean analytifications of $\mathbb{T}$ and X_Σ.
- Tropicalization map

$$\mathrm{trop}\colon \mathbb{T}^{\mathrm{an}} \longrightarrow N_{\mathbb{R}} = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{R}),\ \ |\ | \longmapsto (m \mapsto -\log|\chi_m|)$$

extends (Kajiwara/Payne) to continuous proper surjective map

$$\mathrm{trop}_\Sigma\colon X_\Sigma^{\mathrm{an}} \longrightarrow N_\Sigma.$$

- Maximal affinoid torus $\mathbb{S} := \mathrm{trop}^{-1}(\{0\})$ of $\mathbb{T}^{\mathrm{an}}$ acts on X_Σ^{an}.
- trop describes orbit space for this action.
- **Extended moment map** $\psi : W \to X_\Sigma^{\mathrm{an}}$ induces

$$\psi_{\mathrm{trop}} : W \longrightarrow N_\Sigma.$$

- In his 2016 thesis, Philipp Jell used extended tropicalizations to **tropicalize smooth (p,q)-forms on X^{an} globally**.
- In order to do the same for Mihatsch's δ-forms on X^{an}, one has to develop on **tropical toric varieties** the theory of Lagerberg forms, polyhedra, tropical cycles and divisors, polyhedral currents, δ-forms, Green currents, etc...
- Then one can show the following **global tropicalization result**:

Proposition (Burgos–Gubler–K. 2025)

Let X be a quasi-projective variety over K and let β be a δ-form on X^{an} with compact support. Then there exist an immersion $\psi\colon X \to \mathbb{P}^n$ and a δ-form β_{trop} on $\mathrm{Trop}(\mathbb{P}^n)$ such that $\beta = \psi^*_{\mathrm{trop}}(\beta_{\mathrm{trop}})$.

- **Aim.** Develop Arakelov theory on tropical toric variety N_Σ.

- **Jell, Shaw, Smacka Adv. Geom. 2019.** Get good extension of sheaves of Lagerberg forms and currents from $N_{\mathbb{R}}$ to N_Σ by asking that the forms become constant along the boundary w.r.t. the maps $\pi_{\sigma,\tau}\colon N(\tau) \to N(\sigma)$ if $\tau \prec \sigma$ is face.
- Lagerberg forms of top degree on N_Σ vanish at boundary.
- A **polyhedron of sedentarity** $\tau \in \Sigma$ **in** N_Σ is by definition the closure in N_Σ of a classical polyhedron in $N(\tau)$.
- A polyhedron in N_Σ is **constant along the boundary**, if it is constant along the boundary w.r.t. the maps $\pi_{\sigma,\tau}\colon N(\tau) \to N(\sigma)$ if $\tau \prec \sigma$ is face.
- Polyhedra in N_Σ that are constant along the boundary behave less pathological.
- **Burgos–Gubler-K. 2025.** Extend sheaves of δ-forms of Mihatsch from $N_{\mathbb{R}}$ to N_Σ by asking that polyhedra involved in polyhedral currents are of sedentarity (0) and constant along boundary.

- **Tropical Cartier divisor D on N_Σ** is given by finite family $D = (U_i, \phi_i)_{i\in I}$ with $(U_i)_{i\in I}$ open covering of N_Σ and $\phi_i : N_\mathbb{R} \to \mathbb{R}$ piecewise affine such that for every $i, j \in I$, the difference $\phi_i - \phi_j : U_i \cap U_j \cap N_\mathbb{R} \to \mathbb{R}$ is affine and extends to continuous function $U_i \cap U_j \to \mathbb{R}$.
- **Intersection with tropical Cartier divisor.** If D is a tropical Cartier divisor and $C \in \mathrm{TZ}_d(N_\Sigma)$ is tropical cycle of dimension d of sedentarity (0), then can define $D.C \in \mathrm{TZ}_{d-1}(N_\Sigma)$.
- A **tropical Green function on** N_Σ is a function $g\colon N_\mathbb{R} \to \mathbb{R}$ such that for each $x \in N_\Sigma$ there exist open neighbourhood Ω of x in N_Σ, $m \in M$ and a continuous function $h\colon \Omega \to \mathbb{R}$ such that $g = m + h$ holds on $N_\mathbb{R} \cap \Omega$.
- Call g **of smooth (resp. ps) type** if h can be chosen smooth (resp. ps).

- Given tropical Green function g on N_Σ, there is unique tropical **toric** Cartier divisor D on N_Σ such that if $\sigma \in \Sigma$ and D given on $N_{\langle\sigma\rangle}$ by $m_\sigma \in M$, then $g - m_\sigma$ extends continuously $N_{\langle\sigma\rangle} \to \mathbb{R}$.
- 1**-st Chern** δ**-form.** Let g be a Green function of ps type for tropical toric Cartier divisor D. Define

$$c_1(D,g) \in B^{1,1}(N_\Sigma)$$

as follows: Suppose $x \in N(\sigma)$ and D is given in neighbourhood Ω of x in N_Σ by $m_\sigma \in M$, then there is ps function h on Ω such that $g = m_\sigma + h$ on $\Omega \cap N_\mathbb{R}$ and

$$c_1(D,g)|_\Omega := d'd''h.$$

- **Tropical Poincaré–Lelong equation.** Let g be Green function of ps type for the tropical toric Cartier divisor D and C tropical cycle of sedentarity (0) on N_Σ. Then have identity of δ-currents on N_Σ

$$d'd''(g \wedge \delta_C) = c_1(D,g) \wedge \delta_C - \delta_{D\cdot C}.$$

- Given a pair $(C,T) \in \mathrm{TZ}^p(N_\Sigma) \times E^{p-1,p-1}(N_\Sigma)$, we define

$$\omega(C,T) := d'd''T + \delta_C \in E^{p,p}(N_\Sigma).$$

- If the formal δ-current $\omega(C,T)$ is induced by a δ-form, then we call T **a Green δ-current for C.**
- **Intersection with arithmetic divisors (D,g).** For ps Green function g for D on N_Σ and $(C,T) \in \mathrm{TZ}^p(N_\Sigma) \times E^{p-1,p-1}(N_\Sigma)$ such that D intersects C properly, let

$$(D,g) * (C,T) := (D \cdot C, g \wedge \delta_C + \omega(D,g) \wedge T)$$

  in $\mathrm{TZ}^{p+1}(N_\Sigma) \times E^{p,p}(N_\Sigma)$.
- Get commutativity, projection formula and

$$\omega\big((D,g) * (C,T)\big) = \omega(D,g) \wedge \omega(C,T).$$

Global tropicalization of Cartier divisors and Green functions.

- Let D be Cartier divisor on X. If there is morphism $\psi\colon X \to X_\Sigma$ and toric Cartier divisor E of X_Σ which intersects $\psi(X)$ properly with $\psi^*(E) = D$, then D **is tropicalized by** ψ **and** D_{trop} when D_{trop} is tropical toric Cartier divisor on N_Σ corresponding to E.
- Let g be a Green function of ps type for Cartier divisor D on X. We say that g is **tropicalized by a morphism** $\psi\colon X \to X_\Sigma$ **and by** g_{trop} **on** N_Σ if Cartier divisor D is tropicalized by ψ and tropical toric Cartier divisor D_{trop} on N_Σ and if there is tropical Green function g_{trop} of ps type for D_{trop} on N_Σ such that

$$g = g_{\mathrm{trop}} \circ \psi_{\mathrm{trop}} : \psi^{-1}(\mathbb{T})^{\mathrm{an}} \xrightarrow{\psi_{\mathrm{trop}}} N_{\mathbb{R}} \xrightarrow{g_{\mathrm{trop}}} \mathbb{R}.$$

Theorem (Burgos–Gubler–K. 2025)

Let $\widehat{D_0} = (D_0, g_0), \dots, \widehat{D_d} = (D_d, g_d)$ be arithmetic divisors on the d-dimensional projective variety X with Green functions g_j of ps type such that $D_0, \dots, D_d$ intersect properly.

- *There exist a complete fan Σ and a closed immersion $\psi\colon X \to X_\Sigma$ into the proper toric variety X_Σ which tropicalizes g_j for every $j = 0, \dots, d$ by a Green function $g_{j,\mathrm{trop}}$ of ps type for a tropical toric Cartier divisor $D_{j,\mathrm{trop}}$ on N_Σ.*
- *For the tropical cycle $C := \psi_{\mathrm{trop}}(X)$, we have the formula*

$$\lambda_{\widehat{D_0},\dots,\widehat{D_d}}(X) = ((D_{0,\mathrm{trop}}, g_{0,\mathrm{trop}}) * \cdots * (D_{d,\mathrm{trop}}, g_{d,\mathrm{trop}}) * (C, 0))(1)$$

that computes the local height purely on N_Σ.

Thank you for your attention!